

Trader Performance vs. Market Sentiment

A Study of Bitcoin Fear & Greed Index vs. Hyperliquid Historical Execution Data

1. Executive Summary

This report analyzes how cryptocurrency market sentiment - measured by the Bitcoin Fear & Greed Index - influences trader behavior and profitability. Utilizing historical trading data from Hyperliquid and daily sentiment indices, this study evaluates the correlation between market psychology and trade execution metrics. The analysis reveals that market sentiment significantly drives trade sizes and profitability, with fearful markets offering high-conviction, large-scale profit opportunities, and extreme greed leading to high win-rate but capital-diluted outcomes.

2. Methodology

- Data Loading & Inspection: Loaded historical Hyperliquid trade records and the Fear & Greed Index dataset, verifying dimensions, schema, and completeness.
- Cleaning & Conversion: Corrected invalid epoch timestamps to standard datetime formats and aligned timezones.
- Alignment & Merging: Extracted daily date components and merged both datasets on daily dates.
- Feature Engineering: Formulated key performance indicators (KPIs) at both the trader and aggregate daily levels.
- Exploratory Analysis (EDA): Grouped execution metrics by sentiment categories and visualized distributions.
- Trader Segmentation: Segmented traders by frequency and consistency of returns to isolate behavior.
- Visuals & Recommendations: Created statistical plots to extract business insights and define actionable trading rules.

3. Data Preparation

- Quality Checks: Checked missing values, duplicate records, and verified column data types.
- Timestamp Standardization: Timestamp IST was converted to datetime because the Timestamp column contained incorrect epoch values.
- Merging Alignment: Date columns were extracted for merging. After merging on daily dates, only 6 records had missing sentiment, which is negligible.

4. Features Engineered

- Daily PnL per Trader: Realized profit and loss per trader aggregated daily.
- Win Rate & Avg Size: Percentage of profitable closed trades and average dollar value of positions.
- Activity & Bias: Number of trades per day and Long vs. Short ratio to assess directional bias.
- Omitted Metrics: Leverage analysis was not performed because the dataset lacked leverage columns.

5. Trader Segmentation

Traders were segmented into Frequent vs. Infrequent Traders and Consistent Winners vs. Other Traders. Segmenting behavior allows us to isolate behavioral patterns, determine if performance is driven by a small cohort of high-performing traders, and understand risk-management styles under different market regimes.

6. Exploratory Data Analysis

Analysis was performed across sentiment categories: Extreme Fear, Fear, Neutral, Greed, and Extreme Greed. Six key charts were created to evaluate trading dynamics: Average Closed PnL, Win Rate, Average Trade Size, Number of Trades, Long vs Short Bias, and Closed PnL Distribution (Boxplot).

7. Key Findings

Sentiment	Win %	Avg Size	Mean PnL	Total PnL
Extreme Greed	46.49%	\$3,112.25	\$67.89	\$2,715,171
Greed	38.48%	\$5,736.88	\$42.74	\$2,150,129
Neutral	39.70%	\$4,782.73	\$34.31	\$1,292,921
Fear	42.08%	\$7,816.11	\$54.29	\$3,357,155
Extreme Fear	37.06%	\$5,349.73	\$34.54	\$739,110

- High-Conviction Sizing in Fear: Despite a moderate win rate (42.08%), the Fear regime generated the highest Total PnL (\$3.36M) and largest Avg Trade Size (\$7,816.11). This suggests traders deploy larger, higher-conviction positions during fear, buying major support/capitulations for substantial returns.
- Extreme Greed Over-Trading: Extreme Greed yielded the highest Win Rate (46.49%) and Mean PnL (\$67.89), but the lowest Avg Trade Size (\$3,112.25). This shows that while rising markets make winning easier, FOMO drives traders to over-diversify and dilute capital into smaller, fragmented positions.
- Capital Erosion in Neutral/Extreme Fear: Neutral and Extreme Fear are high-risk, low-reward. Neutral has the lowest Mean PnL (\$34.31), reflecting directionless churn. Extreme Fear has the lowest Win Rate (37.06%) and low Mean PnL (\$34.54), leading to the lowest Total PnL (\$0.74M), showing that large sizes (\$5,349.73) in panic markets destroy capital.

8. Actionable Recommendations

- Strategic Capital Allocation: Focus capital on high-probability setups during Fear (for discount plays) and Extreme Greed (momentum), enforcing strict profit-taking during greed.
- Risk De-escalation Rules: Programmatically reduce exposure (lower position limits, tighter stops) during Neutral and Extreme Fear markets to prevent capital erosion.
- Systematic Position Sizing: Adjust position sizes dynamically based on historical regime profitability rather than emotional impulse or market FOMO.

9. Limitations

- Absence of Leverage: Leverage details were not present in the dataset, preventing analysis of liquidations or true risk-adjusted returns.
- Drawdown Approximation: Portfolio equity history was unavailable, meaning drawdowns and maximum risk-exposure metrics are only estimates.
- Historical Generalizability: Findings reflect historical correlations that may shift under changing macroeconomic conditions or structural market changes.

10. Conclusion

Market sentiment is a powerful leading indicator of trader behavior. By quantifying how emotional extremes affect sizing and win rates, traders and risk managers can build objective, sentiment-based frameworks to optimize position sizing, manage platform risk, and avoid cognitive biases that lead to capital erosion.